

# Overview

Part 04 —  Large Language Models · 04-llm/README.md

## Module 04 — Large Language Models

**Where you are:** Stop 4 of 16. **Time:** ~25 hours · **Prereq:** Module 03 [03-nlp] — non-negotiable.

A large language model is the Module 03 transformer, scaled, trained on most of the public internet, and then bent into a helpful assistant by post-training. This module covers the three phases of that pipeline (pretraining → alignment → inference), the decoding knobs you actually control, and the failure modes you will spend your career designing around.

### Learning objectives

1. Describe the full lifecycle: pretraining → SFT → preference optimisation (RLHF/DPO) → deployment.
2. Use **scaling laws** to reason about compute, data and parameter trade-offs — and explain what Chinchilla changed.
3. Control generation deliberately: temperature, top-k, top-p, min-p, repetition penalty, and structured decoding.
4. Choose correctly between **prompting** → **RAG** → **fine-tuning** → **continued pretraining**, and defend the choice on cost.
5. Explain quantization, LoRA/QLoRA, distillation, and speculative decoding well enough to size a deployment.
6. Name the real failure modes — hallucination, sycophancy, context rot, prompt sensitivity — and mitigate each.

### Suggested path

| # | Do this                   | File                                 | Time |
|---|---------------------------|--------------------------------------|------|
| 1 | How an LLM is built       | notes/01-how-llms-are-built.md       | 5h   |
| 2 | Using them well           | notes/02-inference-and-adaptation.md | 5h   |
| 3 | Sampling, felt not read   | code/sampling_strategies.py          | 2h   |
| 4 | Cost & capacity model     | code/llm_cost_calculator.py          | 2h   |
| 5 | Prompt patterns that work | code/prompt_patterns.md              | 2h   |
| 6 | Slides                    | slides/                              | 1h   |
| 7 | Lab                       | lab/EXERCISES.md                     | 6h   |
| 8 | Papers                    | papers/PAPERS.md                     | 5h   |

## The 16 terms you must own

pretraining · SFT · RLHF · DPO · scaling law · context window · temperature · top-p · logits · perplexity · in-context learning · chain-of-thought · LoRA · quantization · speculative decoding · hallucination

### Exit check

Given a use case, you can produce a defensible one-page recommendation: which model, prompted or fine-tuned, at what context length, what it will cost per 1,000 requests, what its top-3 failure modes are, and how you'd measure them.